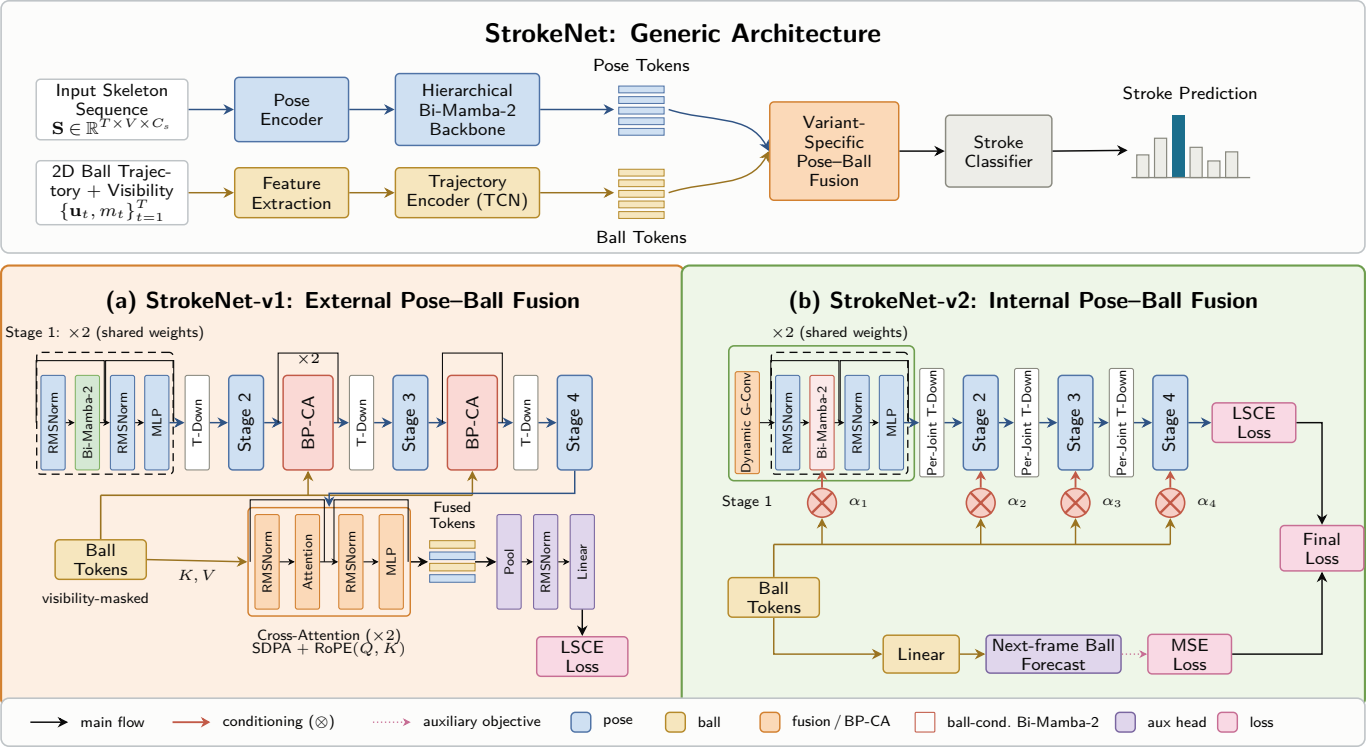
StrokeNet: Generic Architecture
Input Skeleton Sequence $\mathbf{S} \in \mathbb{R}^{T \times V \times C_s}$
Pose Encoder
Hierarchical Bi-Mamba-2 Backbone
Pose Tokens
2D Ball Trajectory + Visibility $\{\mathbf{u}_t, m_t\}_{t=1}^{T}$
Feature Extraction
Trajectory Encoder (TCN)
Ball Tokens
Variant-Specific Pose–Ball Fusion
Stroke Classifier
Stroke Prediction
(a) StrokeNet-v1: External Pose–Ball Fusion
Stage 1: ×2 (shared weights)
RMSNorm
Bi-Mamba-2
RMSNorm
MLP
T-Down
Stage 2
×2
BP-CA
T-Down
Stage 3
BP-CA
T-Down
Stage 4
Ball Tokens
visibility-masked
K, V
RMSNorm
Attention
RMSNorm
MLP
Fused Tokens
Pool
RMSNorm
Linear
Cross-Attention (×2)
SDPA + RoPE(Q, K)
LSCE Loss
(b) StrokeNet-v2: Internal Pose–Ball Fusion
×2 (shared weights)
Dynamic G-Conv
RMSNorm
Bi-Mamba-2
RMSNorm
MLP
Per-Joint T-Down
Stage 2
Per-Joint T-Down
Stage 3
Per-Joint T-Down
Stage 4
LSCE Loss
Stage 1
α_1
α_2
α_3
α_4
Final Loss
Ball Tokens
Linear
Next-frame Ball Forecast
MSE Loss
main flow
conditioning (⊗)
auxiliary objective
pose
ball
fusion / BP-CA
ball-cond. Bi-Mamba-2
aux head
loss